

자동제어시스템(공) (MECH309-01)

학기	2026년 1학기
담당교수	박석호
이수구분	공통필수
학점	3.0
강의시간	화10:30-12:00(E7 - 236), 목10:30-12:00(E7 - 236)
강의실	E5 511 (교수실)
소속	기초학부

교과개요

This course introduces the dynamic modeling and design of automatic control systems as applied to a variety of systems: including not only mechanical and electrical control systems but also chemical reaction process control and biological response and adjusting systems. Students will study characteristics by transfer function between input and output for the automatic control systems. Topics include the properties and advantages of feedback control (e.g., PID) control systems, time-domain and frequency-domain performance measures, stability and degree of stability, the Root Locus method, Bode diagram, Nyquist criterion, frequency-domain design analysis.

주차별 강의계획

1주차: Overview and Important Concepts
2주차: Modeling of Dynamics I: Mechanical and Electrical Systems
3주차: Modeling of Dynamics II: Electro-Mechanical Systems
4주차: Feedback Control & PID Control Design
5주차: Feedback Control & PID Control Design
6주차: PID Control & Applications
7주차: PID Control Tuning: Ziegler-Nichols Method
8주차: Midterm Exam
9주차: Controller Design I: Root Locus Method
10주차: Controller Design II: Pole Placement
11주차: System Response and Time Domain Analysis
12주차: Frequency Response Analysis
13주차: Stability Analysis
14주차: Case Studies: Control System Designs & Applications
15주차: Review
16주차: Final Exam

수업방법

Lecture